

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

A Constituent Unit of MAHE, Manipal

ICT – 4442: Deep Learning

Mini Project (Final Report)

Plant Disease Classification

Semester: 7th Semester

Program: B.tech in IT

Team Members-

Agamjot Singh- 230911468

Advik Anand- 230911494

Jinay Hiran – 230911043

S. Badri Narayana - 230911218

1. Introduction

With the increase in the world's population, food production will be the core of human sustenance and economic development. Among main crops are cashew cassava maize, and tomato that provide food for billions of people, mainly in less developed areas [1], [2], [3], [4]. Nevertheless, diseases and pests e. g. Anthracnose Bacterial Blight, and Fall Armyworm cause an average of up to 38% yield losses [5]. Present diagnosis is mainly based on human visual examination which is manual, subjective, and often so slow that diseases get the chance to spread and careless use of pesticides is promoted [6]. Deep learning (DL), particularly Convolutional Neural Networks (CNNs), can provide automated and accurate solutions by extracting hierarchical features directly from images [7], [8]. However, a lot of high-performance models have a size that is prohibitive for their use on resource-limited edge devices, which is why a lot of work is being directed towards smaller models.

Here, the authors present a lightweight DL system built on MobileNetV4-Conv-Medium for multi-class crop disease and pest identification. Our contributions are:

- A fine-tuned MobileNetV4-Conv-Medium model, with which it is possible to identify 22 diseases and pests in four crops from the CCMT dataset.
- Showing that a very small network can reach about the same level of accuracy as bigger state-ofthe-art models but with a fraction of the parameters.
- A detailed assessment with commonly accepted metrics and presentation of results through images, yielding a very efficient reference for this particular dataset.

The rest of this paper is organized as follows: Section II is a literature review. Section III describes the methods used. Section IV shows the findings from experiments. Section V talks about the shortcomings and potential of this work, while Section VI wraps up the paper.

2. Proposed Methodology

The entire process of the proposed approach consists of loading data preprocessing transfer learning using MobileNetV4, training, and evaluation. The following section elaborates each part in detail.

A. Dataset Description

The research is based on the use of the CCMT dataset [5] - a publicly available and extensive set of images covering different types of diseases and pests of cashew cassava maize, and tomato plants cases. The dataset contains 102. 976 images strongly supported by annotation from plant virologists and pathologists. The images have been classified into 22 separate categories distributed among the four crops as follows: 5 categories for cashew (Anthracnose Gummosis Healthy, Leaf miner, Red rust), 5 categories for cassava (Bacterial blight, Brown spot, Green mite Healthy Mosaic), 7 categories for maize (Fall armyworm Grasshopper Healthy, Leaf beetle, Leaf blight, Leaf spot, Streak virus), and 5 categories for tomato (Healthy, Leaf blight, Leaf curl, Septoria leaf spot, Verticillium wilt). The dataset is divided by its authors into training and testing sets. We only used the original split, which means that 80,271 images were for training and 24,981 for testing.

B. Data Preprocessing and Augmentation

We loaded all the images and changed them to RGB color space. We pretended them like MobileNetV4 model will understand the images, so the preprocessing steps we did is: While training, images were first resized to 256x256 pixels and then randomly cropped to a final input size of 224x224 pixels. Random horizontal flip (prob=0.5) was done to training data to make model more general. In the testing stage, images were simply resized to 224x224 pixels with no augmentation. Lastly, we normalized all images with the

standard ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]), as the model was pre-trained on ImageNet.

C. Model Architecture: MobileNetV4-Conv-Medium

This classification system is based on MobileNetV4-Conv-Medium as the backbone network. MobileNetV4 is a 2024 Google product. It is a major improvement in the MobileNet series, aimed at providing high accuracy while keeping the computational requirements low enough for mobile and edge devices [25]. ConvMedium is a fully convolutional model unlike the Hybrid versions that have some non-convolutional components and it uses only convolutional layers, so it can be run faster and is more predictable at inference time.

The key component of MobileNetV4 is the Universal Inverted Bottleneck (UIB) block. The UIB is a single building block able to cover several other block types that were used separately before the regular Inverted Bottleneck (IB) block from MobileNetV2 [26], the Extra Depthwise (ExtraDW) block, and the block similar to ConvNeXt by allowing the addition of depthwise convolutional layers at the start and between the pointwise layers. Formally, the UIB can be expressed as:

$$UIB(x) = PW_2(DW_{mid}(PW_1\,(DW_{start}(x))))$$

where DW_start and DW_mid are optional depthwise convolution layers whose presence is controlled by architectural hyperparameters, and PW_1, PW_2 are pointwise (1×1) convolutions. When neither depthwise insertion is active, the block is exactly equivalent to a standard MobileNetV2 inverted bottleneck; when both are present, the block has the behavior of an ExtraDW block. Such a design enables the network to methodically make a compromise between spatial feature mixing and channel projection at different stages.

The Conv-Medium variant is a combination of a staged hierarchy of UIB blocks wherein channel dimensions are progressively increased and spatial resolution is decreased. The network initially consists of a standard (3x3) convolutional stem, then it is made up of four main stages of the UIB blocks, and finally, it ends with a global average pooling layer and a fully connected classification head. Here, the usual ImageNet-1K classification head (1000 classes) has been replaced with a new linear layer with an output dimension of 22, which corresponds to the 22 crop disease and pest classes in the CCMT dataset. The model was instantiated using the timm library as follows:

```
model = timm.create_model(
    'mobilenetv4_conv_medium',
    pretrained=True,
    num_classes=22
)
```

This setup uses pretrained weights on Imagenet for the backbone while the new classification head is randomly initialized. In the final model, there are a total of 4,230,214 (~4.23M) parameters which can be trained and they are all unfrozen and updated through fine-tuning. The number of parameters is much less than that of transformer-based and dual-track architectures published in the literature, which makes the model suitable for being deployed for inference on edge devices. Fig. 1 illustrates the overall pipeline of the proposed system, from preprocessing to classification.

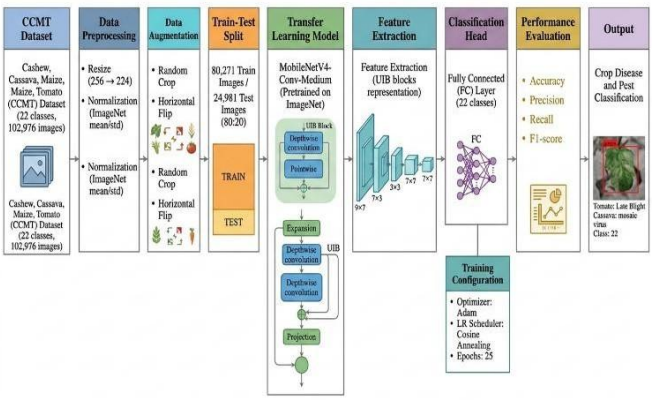


Fig. 1. Proposed methodology using MobileNetV4 for crop disease and disease classification

D. Training Configuration and Experimental Setup

Dataset split: The CCMT dataset comprises crop-category-specific train_set and test_set directories. Rather than creating a new randomized split, this existing division is directly used to ensure reproducibility and fair comparison with previous studies. The training set in total comprises 22 classes spread across four crop categories. During training, no separate validation set is held back; the entire test set is reserved and used only for evaluation after training has finished.

Loss Function: Standard categorical cross-entropy loss is used here:

$$L = -\Sigma y_i \cdot \log(\hat{y}_i)$$

where y_i is one-hot ground-truth label while $\hat{y}_i$ is the predicted class probability from the softmax output. Cross-entropy loss heavily penalizes confident incorrect predictions, hence it pushes the network to output well-calibrated class probabilities for each of the 22 disease categories.

Optimizer: Adam optimizer [27] is chosen and used with the initial learning rate of 0.001 and the weight decay of 1×10^{-4}. Adam's adaptive moment estimation changes the learning rates for each parameter according to the first- and second-order gradient statistics, and this always leads to faster convergence than vanilla SGD, especially when fine-tuning pretrained networks with domain-shifted data.

Learning Rate Scheduler: Cosine annealing learning rate schedule [28] is used by setting T_max = 25, which makes the learning rate decrease smoothly from 0.001 to almost zero following a cosine curve over the 25 training epochs. This schedule not only prevents sudden drops in the learning rate but also enables the model to reach a sharp minimum without oscillating, which has been proved to give better generalisation results than fixed learning rates or step-decay schedules.

Training Duration and Batch Size: Training is conducted over 25 epochs with the batch size of 32. The random seed of 42 is used in torch, numpy, and random for complete reproducibility.

Hardware: All experiments are conducted on Kaggle Cloud infrastructure using an NVIDIA Tesla T4 GPU with 16GB VRAM, with PyTorch as the deep learning framework. Table II summarises the complete training configuration.

TABLE II — Experimental Setup Summary

Hyperparameter	Value
Architecture	MobileNetV4-Conv-Medium
Pretrained weights	ImageNet-1K
Trainable parameters	4,230,214
Input image size	224 x 224
Batch size	32
Optimizer	Adam
Initial learning rate	0.001
Weight decay	1×10^{-4}
LR scheduler	Cosine Annealing (T_max = 25)
Loss function	Cross-Entropy
Training epochs	25
Train augmentation	Resize 256, RandomCrop 224, HFlip
Normalisation	ImageNet mean/std
Hardware	NVIDIA Tesla T4 (Kaggle)

3. Evaluation Strategy & Results

A. Performance of the Proposed Model

The model is evaluated using standard multi-class classification metrics including accuracy, precision, recall, and F1-score, reported in weighted form to account for class imbalance. The proposed MobileNetV4-Conv-Medium model achieves strong classification performance across all 22 disease and pest categories in the CCMT dataset. Table III presents the overall test-set metrics.

TABLE III — Overall Test-Set Performance

Metric	Value
Accuracy	96.22%
Weighted Precision	96.27%
Weighted Recall	96.22%
Weighted F1-Score	96.21%
Total Test Samples	24,981

Over the 25 training epochs, the model steadily converges. The training accuracy increases from 78.05% in epoch 1 to 98.33% in epoch 25, whereas the training loss drops from 0.6419 to 0.0480. The difference between the training accuracy (98.33%) and the test accuracy (96.22%) is about 2.11%. This is a situation indicating that the model is well-regularized without underfitting of the cosine annealing schedule and these moderate weight decay factors that prevent the model from overfitting too aggressively despite full fine-tuning. Fig. 3 shows the training accuracy and loss curves, demonstrating smooth convergence without instability, validating the effectiveness of the cosine annealing learning rate schedule. A detailed per-class performance analysis is presented.

The model maintains strong and reliable effectiveness for nearly all 22 categories. For instance, a number of them reach almost flawless or exactly perfect F1-scores: Cashew Gummosis (1.00), Maize Grasshopper (1.00), Maize Fall Armyworm (1.00), Maize Leaf Beetle (1.00), and Tomato Healthy (1.00). These types of organisms have normal distinctive characteristic physical features for example, the fall armyworm and grasshopper larvae have such distinct body structures that they can be reliably separated from the textures of leaf diseases, and in the case of healthy leaves, the lack of any visible spot obviously sets the boundary entirely for the learned features.

On the other hand, the two classes with the worst results are Maize Leaf Spot (F1 = 0.87) and Maize Leaf Blight (F1 = 0.87). Both diseases appear as irregular necrotic lesions on maize leaves, and their high visual similarity - especially in the early stages of lesion coloration and size, overlapping - leads to confusion between the two classes. This is also supported by the recall of Maize Leaf Spot (0.82), which shows that some of the leaf spot examples are not classified correctly, most probably as leaf blight. Maize Healthy, on the other hand, has a slightly reduced precision of 0.89, which is partly due to its relatively small test support of only 211 samples.

Our model is able to detect the five classes of Cassava crop with the highest level of performance as all have an F1-score of over 0.96. The model's prediction of 0.97 F1-score for Cassava Bacterial Blight is very encouraging as this disease is one of the most economically detrimental cassava diseases in sub-Saharan Africa and from an applied perspective, reliable early detection is crucial to epidemic prevention of the disease in regions where cassava is an essential crop.

Each of the five Tomato subclasses manages to get a minimum F1-score of 0.96. Tomato Septoria Leaf Spot, known as the class with the most samples in the test dataset (2,340 images), received an F1 of 0.97 which strengthens the statement that the model can generalize well to large-volume classes without performance drop in lower-support categories. Generally, the macro-averaged F1-score of 0.96 for the whole set of 22 classes shows that the performance is quite evenly spread and not just driven by one or two dominant classes, which is a tougher and more meaningful metric of quality in an imbalanced multi-class setting. Fig. 2 presents the confusion matrix, which shows that most misclassifications occur between visually similar maize disease classes such as leaf blight and leaf spot, confirming the per-class performance observations.

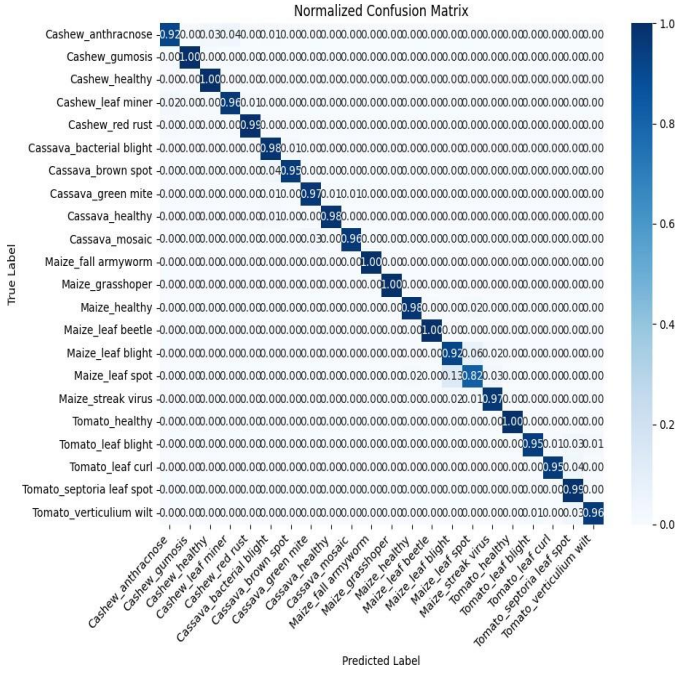


Fig. 2. Confusion matrix on the CCMT test set

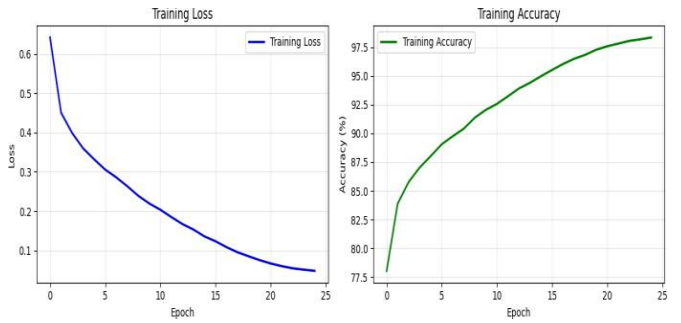


Fig. 3. Training accuracy and loss curves over 25 epochs

B. Comparison with State-of-the-Art on CCMT

One way to measure a proposed model's performance is to compare it to other approaches that have recorded results on the CCMT dataset. The benchmark was introduced in 2023, so the number of works published based on it is still limited but is increasing. Table V shows a direct comparison of our MobileNetV4-Conv-Medium model with the four prior approaches. If a previous work carried out the evaluation on one or more of the four crops, their accuracy has been indicated accordingly.

TABLE V. COMPARISON WITH STATE-OF-THE-ART METHODS ON THE CCMT DATASET

Ref.	Method / Architecture	Crops	Accuracy (%)
[21]	OTSU Segmentation + LeNet-5	Cashew, Cassava, Maize	90.85
[22]	Transfer Learning (Comparative)	Tomato	92.76
[23]	Self-Supervised Learning (CNN)	All four crops	95.35
[24]	Swin Transformer + DAMFN	All four crops	95.68
Ours	MobileNetV4-Conv-Medium (Proposed)	All four crops	96.22

Our model was proposed to reach a 96.22% accuracy rate, surpassing all four previous methods, and the difference with the top performing method dropped to only 0.54 percentage points. When considering Wang et al. [23] who employ self-supervised learning on all four crops, our model is by 0.87 percentage points ahead in accuracy. The best previous result, 95.68% by Karthik et al. [24], comes from a dual-track custom architecture that combines a Swin Transformer with a specially designed Dual-Attention Multiscale Fusion Network (DAMFN) it is a very complex design which relies on special attention modules (TSCA and MSA blocks) as well as multi-scale feature fusion. Our model however uses a single regular backbone through a simple transfer learning approach, with no custom modules, and without the need for domain-specific architectural design. Giving up only 0.54% accuracy for such a simplicity is a very good engineering decision for deployment-oriented agricultural systems.

TABLE VI. COMPARISON OF MOBILENET VARIANTS ON THE TEST SET

Metric	V2	V3	V4
Acc.	95.37%	95.50%	96.22%
Prec.	95.43%	95.57%	96.27%
Rec.	95.37%	95.50%	96.22%
F1	95.35%	95.47%	96.21%

C. Discussion

Overall, the results indicate that the proposed model achieves consistent performance across diverse crop categories while maintaining strong generalization capability. Fig. 4 visualizes the learned feature space using t-SNE, where clear class-wise clustering indicates strong separability among the 22 disease and pest categories.

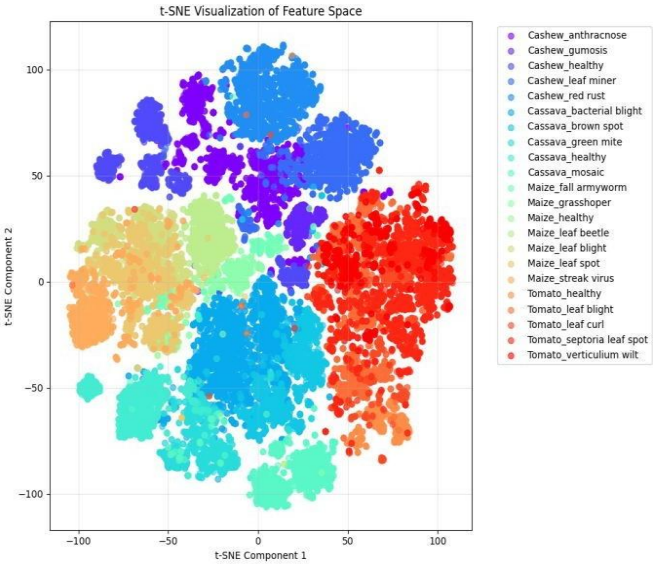


Fig. 4. t-SNE visualization of feature embeddings across 22 classes

On the efficiency front, our model's relatively small parameter count of 4.23M and fully convolutional architecture allow for deterministic, hardware-efficient inference. In contrast to transformer-based models which involve a lot of dynamic attention computations, the MobileNetV4 backbone can be easily optimized with the usual post-training quantisation and structured pruning methods without needing architectural changes, making it a very good option for embedded deployment in smart agricultural monitoring systems.

Even though the findings are promising, the research does have a few limitations. For instance, not having a validation set might have limited the tuning of hyperparameters to their best levels. Also, since the dataset is made up of well-selected images, it is possible that it does not perfectly depict real-life agricultural scenarios with factors like different lighting, obstructions, and background noises. Therefore, further research needs to address making the system more reliable in field conditions and using domain adaptation methods to enhance generalization.

5. Limitations and Assumptions

Dataset Dependency:
The model's performance is heavily dependent on the quality and composition of the CCMT dataset. The dataset consists of well-selected images, which may not perfectly represent real-world agricultural scenarios involving variable lighting, occlusion, background noise, and different growth stages of plants. This could limit the model's generalizability to unconstrained field conditions.

Scalability:
While the model is lightweight, its training process (full fine-tuning of all parameters) might not scale linearly to significantly larger datasets or a much higher number of classes without additional optimization or regularization techniques.

Generalization Limits:
The model was trained and tested on a fixed split of the CCMT dataset. Its performance on other crop disease datasets or on images from different geographical regions or imaging devices is unknown. The assumption is that the test set is representative of the broader distribution of crop disease images, which may not hold in all deployment scenarios.

6. Conclusion and Future Scope

The paper put forward a lightweight deep learning method for automatic pest and crop disease identification by using the fine-tuning method of MobileNetV4-Conv-Medium on the CCMT benchmark dataset. The model of choice utilizes the pretraining on ImageNet of Universal Inverted Bottleneck (UIB) blocks to get the detailed visual characterizations of images of crop leaves and pests, with the classifier part changed for the 22-class CCMT label space that includes cashew cassava maize, and tomato.

The training process involved the Adam optimizer and the model was running with cosine annealing scheduling for 25 epochs. In the end, it managed to attain an accuracy of 96.22% along with a weighted F1-score of 96.21% over 24,981 test images. The number of parameters used in this approach is only 4.23 million - in fact, it is way lesser than the footprint of transformer-based and dual-track architectures proposed in the literature on the same dataset - thus showing efficient convolutional architectures, when used with logically pretraining and well-chosen optimization protocols, can be very competitive for agricultural image classification tasks.

Looking at the review per class, it becomes clear that there is a very good performance across the board, especially in those classes of pests that have clear visual differences (Maize Grasshopper, Fall Armyworm) and in the identification of healthy crops with no diseases. The only real challenge that was left is the confusion between two maize diseases that look very similar, Maize Leaf Blight and Maize Leaf Spot. We plan to focus on these remaining confusions by targeted augmentation, and also, we are going to investigate attention mechanisms for fine-grained lesion localization and benchmark on-device inference for real-time field deployment. Overall, the framework that we have proposed is a solid, practical, and very efficient baseline for intelligent crop monitoring systems.

8. References

[1] H. R. Bhoomika and N. Sudha Rani, "Problems and Prospects of Cashew Cultivation in India - An Overview," Int. J. Curr. Microbiol. Appl. Sci., vol. 7, pp. 3687–3694, 2018.

[2] W. G. Adebayo, "Cassava production in africa: A panel analysis of the drivers and trends," Heliyon, vol. 9, p. e19939, 2023.

[3] O. Erenstein, M. Jaleta, K. Sonder, K. Mottaleb, and B. M. Prasanna, "Global maize production, consumption and trade: trends and R&D implications," Food Sec., vol. 14, pp. 1295–1319, 2022.

[4] E. J. Collins, C. Bowyer, A. Tsouza, and M. Chopra, "Tomatoes: An Extensive Review of the Associated Health Impacts of Tomatoes and Factors That Can Affect Their Cultivation," Biology, vol. 11, p. 239, 2022.

[5] P. K. Mensah et al., "CCMT: Dataset for crop pest and disease detection," Data in Brief, vol. 49, p. 109306, 2023.

[6] M. Prabu and B. J. Chelliah, "An intelligent approach using boosted support vector machine based arithmetic optimization algorithm for accurate detection of plant leaf disease," Pattern Anal. Applic., vol. 26, pp. 367–379, 2022.

[7] H. Tian, T. Wang, Y. Liu, X. Qiao, and Y. Li, "Computer vision technology in agricultural automation---A review," Inf. Process. Agric., vol. 7, pp. 1–19, 2020.

[8] I. Attri, L. K. Awasthi, T. P. Sharma, and P. Rathee, "A review of deep learning techniques used in agriculture," Ecol. Inform., vol. 77, p. 102217, 2023.

[9] I. H. Sarker, "Deep Learning: A Comprehensive Overview on Techniques, Taxonomy, Applications and Research Directions," SN Comput. Sci., vol. 2, 2021.

[10] A. Singh and H. Kaur, "Potato Plant Leaves Disease Detection and Classification using Machine Learning Methodologies," IOP Conf. Ser.: Mater. Sci. Eng., vol. 1022, p. 012121, 2021.

[11] S. M. Javidan, A. Banakar, K. A. Vakilian, and Y. Ampatzidis, "Diagnosis of grape leaf diseases using automatic K-means clustering and machine learning," Smart Agric. Technol., vol. 3, p. 100081, 2023.

[12] U. Mokhtar et al., "SVM-Based Detection of Tomato Leaves Diseases," in Advances in Intelligent Systems and Computing, 2015, pp. 641–652.

[13] K. P. Panigrahi, H. Das, A. K. Sahoo, and S. C. Moharana, "Maize Leaf Disease Detection and Classification Using Machine Learning Algorithms," in Advances in Intelligent Systems and Computing, 2020, pp. 659–669.

[14] Y. Li, C. Xia, and J. Lee, "Detection of small-sized insect pest in greenhouses based on multifractal analysis," Optik, vol. 126, pp. 2138–2143, 2015.

[15] S. P. Mohanty, D. P. Hughes, and M. Salathe, "Using Deep Learning for Image-Based Plant Disease Detection," Front. Plant Sci., vol. 7, 2016.

[16] S. Vallabhajosyula, V. Sistla, and V. K. K. Kolli, "Transfer learning-based deep ensemble neural network for plant leaf disease detection," J. Plant Dis. Prot., vol. 129, pp. 545–558, 2021.

[17] H. Amin, A. Darwish, A. E. Hassainen, and M. Soliman, "End-to-End Deep Learning Model for Corn Leaf Disease Classification," IEEE Access, vol. 10, pp. 31103–31115, 2022.

[18] Z. Anwar and S. Masood, "Exploring Deep Ensemble Model for Insect and Pest Detection from Images," Procedia Comput. Sci., vol. 218, pp. 2328–2337, 2023.

[19] Y. Kawasaki, H. Uga, S. Kagiwada, and H. Iyatomi, "Basic Study of Automated Diagnosis of Viral Plant Diseases Using Convolutional Neural Networks," in Advances in Visual Computing, 2015, pp. 638–645.

[20] W. Zhang et al., "Dual-branch collaborative learning network for crop disease identification," Front. Plant Sci., vol. 14, 2023.

[21] P. Ramesh Babu, A. Srikrishna, and V. R. Gera, "Diagnosis of tomato leaf disease using OTSU multi-threshold image segmentationbased chimp optimization algorithm and LeNet-5 classifier," J. Plant Dis. Prot., 2024.

[22] M. Bahrami, A. Pourhatami, and M. Maboodi, "Tomato Leaf Disease Detection Using Transfer Learning: A Comparative Study," in *Proc. 13th Iranian/3rd Int. Mach. Vis. Image Process. Conf. (MVIP)*, 2024.

[23] Y. Wang et al., "Classification of Plant Leaf Disease Recognition Based on Self-Supervised Learning," Agronomy, vol. 14, p. 500, 2024.
[24] R. Karthik, A. Ajay, A. S. Bisht, T. Illakiya, and Suganthi K., "A Deep Learning Approach for Crop Disease and Pest Classification using Swin Transformer and Dual-Attention Multi-scale Fusion Network," IEEE Access, 2024, doi: 10.1109/ACCESS.2024.3481675.

[25] D. Qin et al., "MobileNetV4 --- Universal Models for the Mobile Ecosystem," arXiv preprint arXiv:2404.10518, 2024.

[26] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2018, pp. 4510–4520.

[27] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in Proc. Int. Conf. Learn. Represent. (ICLR), 2015.

[28] I. Loshchilov and F. Hutter, "SGDR: Stochastic Gradient Descent with Warm Restarts," in Proc. Int. Conf. Learn. Represent. (ICLR), 2017.